



6419 - Mapping undiscovered country; Cloudy simulations of JWST's unique spectral challenges

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The spectral simulation code Cloudy is widely used across the JWST community, with 123 papers appearing in 2023 citing both Cloudy and JWST. This project aims to continue Cloudy's development in support of the spectroscopic challenges posed by JWST's unique spectral bandpass. Cloudy simulations do the microphysics from first principles and rest on a foundation of basic atomic data; energy levels, radiative transition rates, and collisional rate coefficients. Over 95% of a Cloudy download consists of atomic, molecular, or grain data. The result is a full simulation of a non-equilibrium plasma with essentially no free parameters. The specific aims of this project are to incorporate atomic data sets that have appeared in the time since the last major atomic data revision ten years ago and continue Cloudy's development in support of JWST needs. This project will help Cloudy meet the unique challenges posed by JWST's extraordinary capabilities.

OBSERVING DESCRIPTION